

MTH205 Analysis in one variable

Tutorial 4

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(1) Determine the points of continuity of the following functions.

(a) $f(x) = \frac{x^2+2x+1}{x^2+1}$, for all $x \in \mathbb{R}$.

(c) $h(x) = \frac{\sqrt{1+|\sin(x)|}}{x}$, for all $x \in \mathbb{R} \setminus \{0\}$.

(b) $g(x) = \sqrt{x + \sqrt{x}}$, for all $x \geq 0$.

(d) $\varphi(x) = \cos(\sqrt{1+x^2})$, for all $x \in \mathbb{R}$.

(2) Let $A \subseteq \mathbb{R}$. Give an example of a function $f : A \rightarrow \mathbb{R}$ such that f is discontinuous at every point of A but $|f|$ is continuous on A .

(3) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous on $\mathbb{R}$ satisfying $f\left(\frac{m}{2^n}\right) = 0$ for every $m \in \mathbb{Z}$, $n \in \mathbb{N}$. Then show that $f(x) = 0$ for all $x \in \mathbb{R}$.

(4) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Prove that if f is continuous at some point $x_0 \in \mathbb{R}$ then f is continuous at every point of $\mathbb{R}$.

(5) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous at a point $x_0 \in \mathbb{R}$. If we define $h(x) := \sup\{f(x), g(x)\}$ and $e(x) := \inf\{f(x), g(x)\}$ for $x \in \mathbb{R}$ then show that

(a) $h(x) = \frac{1}{2}(f(x) + g(x)) + \frac{1}{2}|f(x) - g(x)|$, $\forall x \in \mathbb{R}$ and h is continuous at x_0 .

(b) $e(x) = \frac{1}{2}(f(x) + g(x)) - \frac{1}{2}|f(x) - g(x)|$, $\forall x \in \mathbb{R}$ and e is continuous at x_0 .